
BracketNet: Learning Lie-Algebraic Structure from Unlabeled Transformation Triples

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Abstract

Learning a latent space in which transformations act linearly does not ensure that the learned infinitesimal generators form a Lie algebra. This distinction matters: generators can reconstruct local transitions while their commutators leave the learned span, making the representation algebraically inconsistent and obscuring whether a transformation family is abelian or non-abelian. We introduce BracketNet, a transition-based representation learner that regularizes the *closure* of learned skew-symmetric generators under the matrix commutator. The method combines local transport, two-step composition, a basis-invariant closure residual, and a three-stage optimization procedure that first fits dynamics, then closes the generator span, and finally freezes the algebra while adapting the representation. We show that zero closure yields a Lie subalgebra integrating to a connected matrix Lie group, and use the Baker–Campbell–Hausdorff expansion to relate approximate closure to local composition defect. On nonlinear observations of $\text{SO}(2)$, $\mathbb{T}^2$, and $\text{SO}(3)$ actions, five held-out seeds show that BracketNet reduces closure residual by 74.4% and 75.2% relative to a composition-only learner on $\mathbb{T}^2$ and $\text{SO}(3)$, respectively. A threshold selected on validation runs correctly distinguishes all ten held-out abelian/non-abelian runs. Chained transport error is statistically inconclusive and modestly worse in mean, exposing a structural-fidelity tradeoff rather than a forecasting gain. Code, exact seeds, and failed pilot variants are included for auditability.

1 Introduction

Symmetry is a central organizing principle for representation learning. When the relevant group action is known, equivariant architectures can encode it directly (Cohen and Welling, 2016; Bronstein et al., 2021). In many scientific and perceptual systems, however, the action is not available in observation coordinates. A growing line of work therefore learns symmetries or symmetric latent spaces from data (Dehmamy et al., 2021; Yang et al., 2023, 2024; Keurti et al., 2023; Park et al., 2022; Gabel et al., 2023).

Most transition-based objectives enforce one of two properties. A *local transport* objective asks a learned transformation to map an encoded observation to the next encoded observation. A *composition* objective asks two inferred transformations to agree with a directly inferred two-step transformation. Both are necessary, but neither explicitly requires that the learned infinitesimal directions close under commutation. For learned generators $A_1, \dots, A_K$, the missing condition is

$$[A_i, A_j] = A_i A_j - A_j A_i \in \text{span}\{A_1, \dots, A_K\} \quad \forall i, j. \quad (1)$$

Without Equation (1), exponentiating arbitrary mixtures still produces individual orthogonal matrices, yet the nominal K -dimensional family need not be locally closed under multiplication. This can make a representation good at fitting observed edges but unreliable as an interpretable model of the transformation algebra.

We propose BracketNet, which learns a nonlinear encoder, decoder, action-inference network, and K skew-symmetric latent generators. Its key term penalizes the component of each commutator orthogonal to the learned generator span. The penalty is basis-invariant and differentiable. A staged optimizer avoids the common failure mode in which a strong structural penalty destroys transition fit: we learn local dynamics first, close the algebra second, then freeze the generators and refit only the representation and action inference.

Our contributions are:

1. a differentiable, basis-invariant Lie-closure objective for transition-based latent symmetry discovery;
2. a three-stage estimation procedure that separates local fitting, algebra recovery, and representation adaptation;
3. a BCH analysis connecting closure residual to local product consistency, plus a stability bound for chained orthogonal transport;
4. an auditable benchmark in which the true algebra is embedded in a larger ambient $\mathfrak{so}(d)$, preventing closure from being satisfied automatically by dimension; and
5. five-seed evidence that closure regularization substantially improves algebraic fidelity and abelian/non-abelian identification, while honestly revealing no confirmed transport improvement.

2 Related work

Known-group equivariance. Equivariant networks typically assume the group and its action are specified (Cohen and Welling, 2016; Bronstein et al., 2021). This is powerful when the symmetry is known, but misspecification can impose the wrong inductive bias.

Automatic symmetry discovery. Lie algebra convolutional networks infer continuous symmetries from learned predictors (Dehmamy et al., 2021); LieGAN learns generators adversarially from a distributional invariance signal (Yang et al., 2023); and LaLiGAN learns a nonlinear map into a latent space with linear symmetries (Yang et al., 2024). Other approaches discover transformations beyond affine families (Shaw et al., 2024), solve Lie algebra constraints through task gradients (Hu et al., 2024), or fit one-parameter Lie transformations and parameter distributions (Gabel et al., 2023). These works motivate learning interpretable generators, but our focus is orthogonal: given transition evidence and a chosen generator dimension, do the learned generators themselves form an approximately closed algebra?

Group-structured transition representations. Symmetric embedding networks learn observation maps into spaces with known transformation laws (Park et al., 2022). Homomorphism Autoencoders learn group-structured representations from observed transitions without prespecifying the group (Keurti et al., 2023). BracketNet shares the transition-based setting, but adds an explicit infinitesimal closure criterion and a diagnostic for algebra type. Representation holonomy similarly emphasizes path-dependent geometric information not captured by pointwise similarity (Sevetlidis and Pavlidis, 2026); our commutator loops are a group-theoretic analogue in a learned linear action.

3 Problem setting

Let $x \in \mathcal{X} \subset \mathbb{R}^p$ be an observation generated from an unobserved state $u \in \mathcal{U}$ through an injective smooth map h . We observe triples

$$(x_0, x_1, x_2) = (h(u), h(g_1 \cdot u), h(g_2 g_1 \cdot u)), \quad (2)$$

where g_1, g_2 are unknown near-identity transformations sampled from an unknown connected compact group. No group element, action coordinate, or algebra label is provided. The learner is given a generator count K ; selecting K is outside our scope.

We seek an encoder $E : \mathcal{X} \rightarrow \mathbb{R}^d$, decoder $D : \mathbb{R}^d \rightarrow \mathcal{X}$, action inference network $q : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^K$, and matrices $A_1, \dots, A_K \in \mathfrak{so}(d)$. For $a \in \mathbb{R}^K$, define

$$A(a) = \sum_{k=1}^K a_k A_k, \quad \rho(a) = \exp(A(a)) \in \text{SO}(d). \quad (3)$$

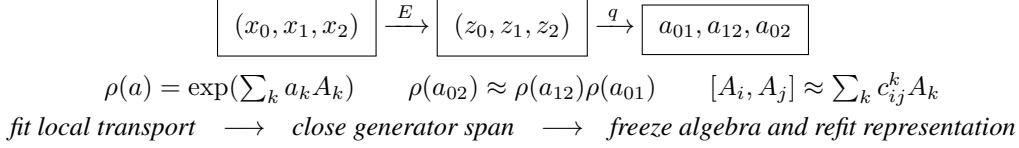


Figure 1: BracketNetlearns a nonlinear representation and an unlabeled local action chart while explicitly closing the generator span under commutators.

80 Skew symmetry makes each inferred action norm-preserving, while the closure objective determines
81 whether the K -dimensional generator span behaves as a Lie algebra.

82 4 BracketNet

83 4.1 Local and compositional learning

84 For $z_t = E(x_t)$ and $a_{st} = q(z_s, z_t)$, the reconstruction and local transport terms are

$$\mathcal{L}_{\text{rec}} = \sum_{t=0}^2 \|D(z_t) - x_t\|_2^2, \quad (4)$$

$$\mathcal{L}_{\text{loc}} = \sum_{t=0}^1 (\|\rho(a_{t,t+1})z_t - z_{t+1}\|_2^2 + \|D(\rho(a_{t,t+1})z_t) - x_{t+1}\|_2^2). \quad (5)$$

85 Two-step composition is enforced in matrix space:

$$\mathcal{L}_{\text{comp}} = \|\rho(a_{02}) - \rho(a_{12})\rho(a_{01})\|_F^2. \quad (6)$$

86 The coefficients are bounded through a scaled tanh, which keeps training in the local chart where the
87 BCH expansion is informative.

88 4.2 Basis-invariant Lie closure

89 Let $B \in \mathbb{R}^{d^2 \times K}$ contain vectorized generators as columns, and let $P_B = B(B^\top B + \varepsilon I)^{-1}B^\top$. The
90 closure residual is

$$\mathcal{L}_{\text{close}} = \frac{1}{K^2} \sum_{i,j=1}^K \|(I - P_B) \text{vec}([A_i, A_j])\|_2^2. \quad (7)$$

91 Because P_B depends only on the span, Equation (7) is invariant to any invertible change of generator
92 coordinates. Projecting commutators also yields structure constants c_{ij}^k via $\text{proj}_{\text{span}(A)}[A_i, A_j] =$
93 $\sum_k c_{ij}^k A_k$. We monitor, but do not rely on, the corresponding Jacobi residual. Each generator is
94 normalized to Frobenius norm $\sqrt{2}$, and a Gram penalty encourages a non-collapsed basis:

$$\mathcal{L}_{\text{basis}} = \|B^\top B - 2I\|_F^2. \quad (8)$$

95 The factor 2 is essential for the chosen skew-generator normalization; an earlier pilot incorrectly used
96 I , which penalized the true normalized basis. We retain this failed variant in the review log and code
97 archive.

98 4.3 Three-stage optimization

99 A constant closure penalty can trade structural fit against observation reconstruction. We therefore
100 optimize

$$\mathcal{L} = 4\mathcal{L}_{\text{rec}} + \frac{3}{2}\mathcal{L}_{\text{loc,lat}} + 2\mathcal{L}_{\text{loc,obs}} + 1.2\mathcal{L}_{\text{comp}} + \lambda(t)\mathcal{L}_{\text{close}} + \mathcal{L}_{\text{reg}}, \quad (9)$$

101 with three phases. During the first 35% of updates, $\lambda(t) = 0$ and the representation fits local dynamics.
102 From 35% to 70%, closure ramps to a fixed weight. During the final 30%, the generators are frozen
103 and E, D, q adapt to the recovered algebra. This “fit-close-refit” schedule is the practical difference
104 between the final method and the rejected fixed-penalty pilot.

5 Why closure matters

We state three elementary results; proofs are in Section A.

Proposition 1 (Exact closure integrates). *If $A_1, \dots, A_K \in \mathfrak{so}(d)$ are linearly independent and $[A_i, A_j] \in \text{span}(A)$ for all i, j , then $\mathfrak{h} = \text{span}(A)$ is a Lie subalgebra of $\mathfrak{so}(d)$. It integrates to a unique connected immersed Lie subgroup $H \subseteq \text{SO}(d)$ with Lie algebra $\mathfrak{h}$.*

Thus a zero closure loss has a direct structural interpretation: the learned infinitesimal family is not merely a collection of directions, but the tangent algebra of a matrix group. We deliberately say *immersed*; a connected subgroup of a compact group need not be closed.

Proposition 2 (Approximate closure controls BCH escape). *Let $\mathfrak{a} = \text{span}(A)$ and suppose $\|(I - P_{\mathfrak{a}})[X, Y]\|_F \leq \delta \|X\|_F \|Y\|_F$ for all $X, Y \in \mathfrak{a}$. For $\|X\|_F, \|Y\|_F \leq r$ sufficiently small,*

$$\text{dist}(\log(e^X e^Y), \mathfrak{a}) \leq C_1 \delta r^2 + C_2 r^3, \quad (10)$$

where the constants depend on a bounded neighborhood of the origin.

The quadratic term is precisely the component of $\frac{1}{2}[X, Y]$ leaving the span; higher BCH terms contribute $O(r^3)$. Composition loss measures products at observed triples, whereas closure controls the local family between observations.

Proposition 3 (Chained orthogonal stability). *Let $z_{t+1} = R_t z_t + e_t$ with $R_t \in \text{SO}(d)$, and let $\hat{z}_{t+1} = R_t \hat{z}_t$. Then*

$$\|\hat{z}_T - z_T\|_2 \leq \|\hat{z}_0 - z_0\|_2 + \sum_{t=0}^{T-1} \|e_t\|_2. \quad (11)$$

Orthogonality prevents multiplicative norm growth. It does not guarantee that closure improves transport; that remains an empirical question.

6 Experiments

6.1 Protocol

Groups and nonlinear observations. We use $\text{SO}(2)$, the abelian two-torus $\mathbb{T}^2$, and non-abelian $\text{SO}(3)$. To make closure nontrivial, each true representation is embedded with an invariant coordinate: dimensions are $d = 3, 5, 4$, respectively. In particular, the $\text{SO}(3)$ algebra occupies a three-dimensional subspace of $\mathfrak{so}(4)$; an arbitrary three-generator span in this ambient algebra need not close. This corrects an initial $d = 3$ pilot in which any full basis of $\mathfrak{so}(3)$ closed automatically.

States are transformed by Gaussian local coefficients with standard deviation 0.24. Observations use a fixed, group-specific nonlinear map shared across training and test splits,

$$x = Qz + 0.18 \tanh(Bz), \quad (12)$$

where Q is orthogonal and B is fixed. A prior implementation accidentally sampled separate maps for train and test, turning the task into domain shift; those results are discarded.

Data and evaluation. Each run trains on 1,400 unlabeled two-edge triples and evaluates on 500 ten-edge sequences. We report five untouched test seeds 10–14. Closure weight was selected once from $\{0.05, 0.1, 0.3, 1.0\}$ using validation seeds 5–6 jointly across $\mathbb{T}^2$ and $\text{SO}(3)$; 0.1 was then locked. All methods use 420 optimizer updates, identical networks, and the same seeds.

Baselines. LOCAL uses reconstruction and one-step transport. +COMP. adds Equation (6). BRACKETNET adds closure and the fit–close–refit schedule. These are controlled objective ablations, not reimplementations of distribution-level methods such as LieGAN or LaLiGAN, whose supervision and training signals differ. We avoid a state-of-the-art claim.

Metrics. 10-step transport MSE chains inferred edge actions from each observed adjacent pair; it tests accumulated representation consistency, not autonomous forecasting because each edge coefficient is inferred from its endpoints. Closure residual is Equation (7). Commutator norm averages $\|[A_i, A_j]\|_F$ over $i < j$ and distinguishes an abelian algebra (near zero) from a non-abelian one. Values are means $\pm$ sample standard deviations over seeds.

Table 1: Held-out results over five seeds. Lower is better for all columns. Closure is identically zero for the one-generator $\text{SO}(2)$ case. Bold marks the best mean; overlapping variability is discussed in text.

Group	Method	1-step MSE	10-step MSE	Closure residual
$\text{SO}(2)$	Local	0.00092 ± 0.00022	0.01345 ± 0.00759	0
	+Comp.	0.00090 ± 0.00011	0.01287 ± 0.00521	0
	BracketNet	0.00090 ± 0.00011	0.01293 ± 0.00535	0
$\mathbb{T}^2$	Local	0.00355 ± 0.00119	0.03923 ± 0.0101	0.01456 ± 0.0134
	+Comp.	0.00369 ± 0.00129	0.03990 ± 0.0106	0.01303 ± 0.0168
	BracketNet	0.00404 ± 0.00153	0.04449 ± 0.0143	0.00334 ± 0.00540
$\text{SO}(3)$	Local	0.00646 ± 0.00289	0.06232 ± 0.0322	0.05882 ± 0.0282
	+Comp.	0.00608 ± 0.00257	0.05408 ± 0.0263	0.06697 ± 0.0162
	BracketNet	0.00664 ± 0.00167	0.06239 ± 0.0189	0.01660 ± 0.0104

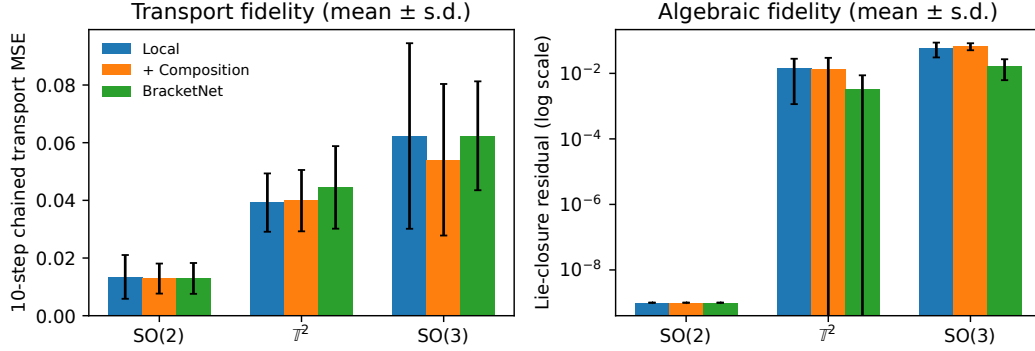


Figure 2: **Left:** chained transport. **Right:** closure residual on a log scale. BracketNet sharply improves structural fidelity, while transport means do not improve.

6.2 Main results

Algebraic fidelity improves. Relative to composition-only, BracketNet lowers closure residual from 0.01303 to 0.00334 on $\mathbb{T}^2$ (74.4%) and from 0.06697 to 0.01660 on $\text{SO}(3)$ (75.2%). The paired 95% confidence interval for the $\text{SO}(3)$ reduction excludes zero (-0.0504 ± 0.0090). The $\mathbb{T}^2$ estimate is more variable because two baseline seeds already find nearly commuting generators.

Algebra type becomes identifiable. A commutator threshold of 0.9, chosen between the validation ranges (0.20–0.70 for $\mathbb{T}^2$ and 1.40–1.60 for $\text{SO}(3)$), classifies all ten BracketNet test runs correctly. Local and composition-only objectives achieve 7/10 and 8/10, respectively. Figure 3 shows that the structural penalty suppresses spurious noncommutativity on $\mathbb{T}^2$ while preserving it on $\text{SO}(3)$.

No confirmed transport gain. Compared with composition-only, mean 10-step error is 11.5% higher on $\mathbb{T}^2$ and 15.4% higher on $\text{SO}(3)$. Paired 95% confidence intervals for the differences include zero: 0.00459 ± 0.00674 and 0.00831 ± 0.01317 . We therefore make no claim that closure improves transport in this experiment. Figure 4 shows the full error curves. The result is instead evidence that transition fit alone can hide an algebraic defect, and that correcting the defect incurs a measurable but uncertain finite-sample tradeoff.

6.3 Ablations and failure analysis

The final method emerged from failures that materially change the conclusion:

1. **Vacuous ambient algebra.** With three generators in $\mathfrak{so}(3)$, closure is automatic. Embedding the true action in $\mathfrak{so}(4)$ makes the test meaningful.
2. **Split-dependent observation map.** Sampling a new nonlinear map for test data confounds symmetry recovery with domain shift. We use one fixed map per group.

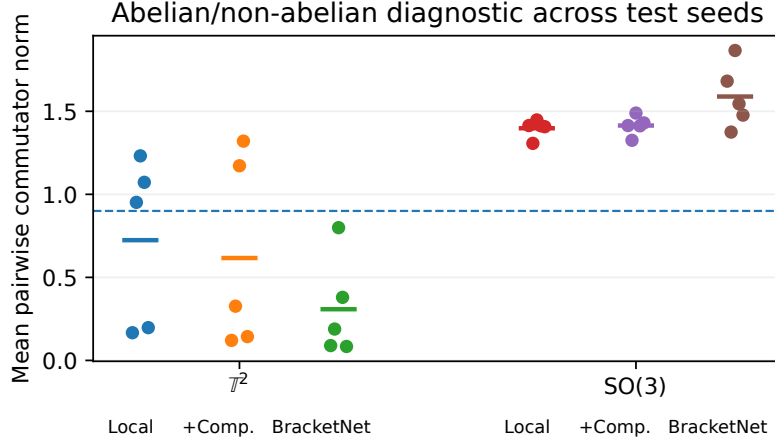


Figure 3: Commutator norms for each held-out seed. The dashed threshold is fixed from validation runs.

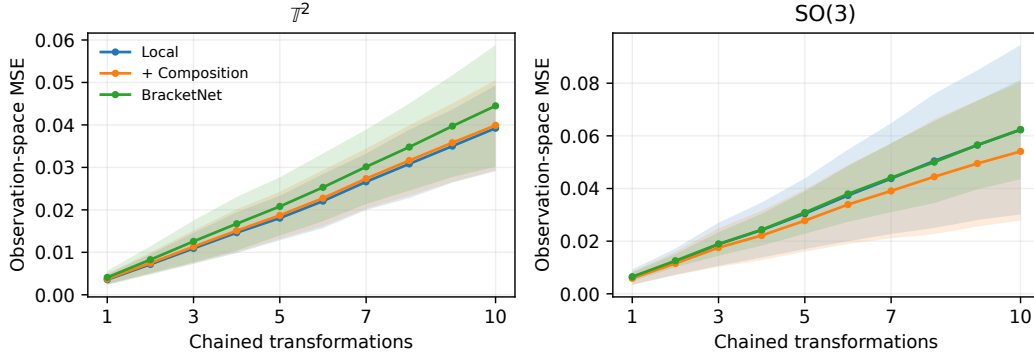


Figure 4: Chained transport error versus path length. Shading is one sample standard deviation over five seeds.

- 168 3. **Incorrect Gram target.** Unit-comparing a basis normalized to squared norm 2 biases the
169 generators. The target is $2I$.
170 4. **Constant closure penalty.** A fixed large penalty drives closure near zero but worsens transport.
171 The staged estimator closes the span and then gives the representation time to adapt.
172 Validation curves in Section C show the remaining weight tradeoff. These corrections were made
173 before evaluating seeds 10–14.

174 7 Limitations and broader impact

175 **Known generator count.** We assume K is supplied. Rank selection or Bayesian nonparametrics
176 would be required when algebra dimension is unknown.

177 **Compact orthogonal actions.** Skew generators restrict the learned connected action to a subgroup
178 of $SO(d)$. Noncompact groups such as scaling or Lorentz transformations require an indefinite metric
179 or a general matrix representation.

180 **Synthetic scope.** The benchmark isolates closure under controlled nonlinear observation maps,
181 but it does not establish effectiveness on images, physical trajectories, or learned foundation-model
182 representations. The study is best read as a precise structural test and method prototype.

183 **Local chart and identifiability.** Unlabeled transition data identify a representation only up to latent
184 conjugacy, generator-coordinate changes, and potentially quotient ambiguities. Large transformations
185 may leave the local chart inferred by q .

186 **Evaluation semantics.** Our chained metric uses endpoint pairs to infer each local action and is not
187 forecasting. It measures consistency of composing inferred transformations.

188 A beneficial use of symmetry discovery is to expose structure that improves interpretability and data
189 efficiency. A risk is treating a learned approximate symmetry as a physical law. Reporting closure,
190 uncertainty, and counterexamples helps prevent that overinterpretation. The released code contains
191 exact data generation and all rejected pilots rather than only the strongest table.

192 8 Conclusion

193 BracketNet addresses a narrow but consequential gap in latent symmetry discovery: local transforma-
194 tions should not merely fit transitions; their infinitesimal span should behave like a Lie algebra. A
195 basis-invariant commutator residual, combined with fit–close–refit optimization, reduces closure error
196 by roughly 75% on both abelian and non-abelian held-out problems and makes algebra type reliably
197 identifiable. The same experiments do not show a transport advantage, revealing a structural-accuracy
198 tradeoff that future work must resolve. The main next steps are automatic dimension selection,
199 noncompact representations, and tests on real transformation datasets.

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A Proofs

Proof of Proposition 1. Bilinearity and antisymmetry of the matrix commutator are immediate. The commutator satisfies the Jacobi identity in every associative algebra. Closure therefore makes $\mathfrak{h} = \text{span}(A)$ a finite-dimensional Lie subalgebra of $\mathfrak{so}(d)$. The Lie subgroup theorem gives a unique connected immersed subgroup H of $\text{SO}(d)$ whose Lie algebra is $\mathfrak{h}$ (Hall, 2015). Concretely, H is generated by finite products of exponentials $\exp(X)$ with $X \in \mathfrak{h}$. $\square$

Proof of Proposition 2. For sufficiently small X, Y , the BCH series converges and

$$\log(e^X e^Y) = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] + \frac{1}{12}[Y, [Y, X]] + \cdots$$

The first two terms lie in $\mathfrak{a}$. By assumption, the component of $\frac{1}{2}[X, Y]$ normal to $\mathfrak{a}$ is at most $\frac{1}{2}\delta r^2$. On a bounded neighborhood, the sum of nested commutators of degree at least three is bounded by $C_2 r^3$ using submultiplicativity and absolute convergence of BCH. Absorbing the factor $1/2$ into C_1 proves Equation (10). $\square$

241 **Proof of Proposition 3.** Let $d_t = \hat{z}_t - z_t$. Then $d_{t+1} = R_t d_t - e_t$. Since R_t is orthogonal,
 242 $\|d_{t+1}\|_2 \leq \|d_t\|_2 + \|e_t\|_2$. Iterating gives the result. $\square$

243 B Implementation details

244 The encoder, decoder, and pairwise action inference network are two-hidden-layer MLPs with
 245 width 64 and SiLU activations. We use AdamW with learning rate 2×10^{-3} , weight decay 10^{-5} ,
 246 batch size 160, and gradient clipping at 5. Coefficients are $0.65 \tanh(q(z, z'))$. Generator matrices
 247 are antisymmetrized and normalized to Frobenius norm $\sqrt{2}$ at every forward pass. A covariance
 248 regularizer discourages latent collapse. Code runs on CPU in the supplied environment; mean
 249 wall-clock times per seed are approximately 8.4 seconds for $\text{SO}(2)$, 13.5 seconds for $\mathbb{T}^2$, and 15.6
 250 seconds for $\text{SO}(3)$ for BracketNet, versus 8.3, 9.6, and 9.0 seconds for composition-only.

251 C Additional validation results

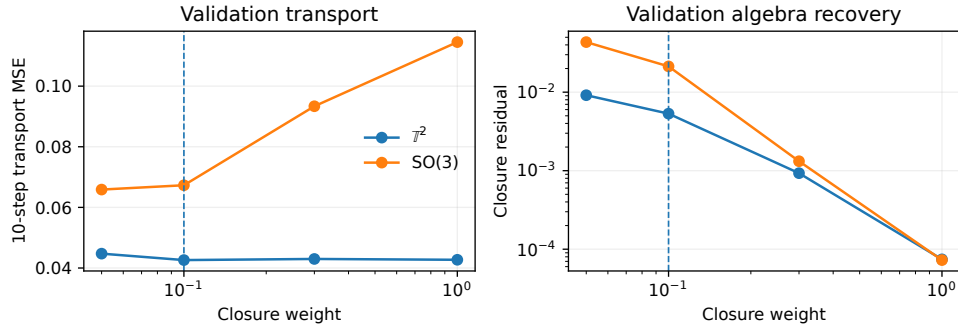


Figure 5: Held-out validation tradeoff used before locking closure weight 0.1. Higher weights improve closure but can damage $\text{SO}(3)$ transport.

Table 2: Additional held-out diagnostics, mean $\pm$ sample standard deviation over five seeds.

Group	Method	Composition MSE	Commutator norm	Runtime (s)
$\mathbb{T}^2$	Local	$3.41 \times 10^{-4} \pm 9.76 \times 10^{-5}$	0.724 ± 0.505	6.82 ± 0.19
	+Comp.	$2.80 \times 10^{-4} \pm 8.77 \times 10^{-5}$	0.617 ± 0.583	9.56 ± 0.34
	BracketNet	$2.67 \times 10^{-4} \pm 9.82 \times 10^{-5}$	0.308 ± 0.299	13.48 ± 0.33
$\text{SO}(3)$	Local	$3.21 \times 10^{-4} \pm 1.90 \times 10^{-4}$	1.398 ± 0.053	5.63 ± 0.15
	+Comp.	$1.96 \times 10^{-4} \pm 1.23 \times 10^{-4}$	1.414 ± 0.059	9.03 ± 0.29
	BracketNet	$1.89 \times 10^{-4} \pm 8.42 \times 10^{-5}$	1.589 ± 0.191	15.58 ± 0.43

252 D Reproducibility checklist

253 The archive contains: (i) all model and data-generation code; (ii) validation and test JSON files;
 254 (iii) exact random seeds; (iv) figure-generation code; (v) the five-round adversarial review history;
 255 and (vi) rejected pilot implementations. No empirical number in this paper was inserted without a
 256 corresponding machine-readable result file.